Evidence budgeting for edge AI under degraded connectivity: a validation study of adaptivity, admissibility, and adversarial metadata, on synthetic and real disaster data

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***Abstract*—Edge AI on degraded field networks must decide what evidence to move, and when to stop waiting. We formulate this as evidence budgeting across concurrent incident windows: a multiple-choice knapsack over a shared byte budget whose latency term prices transfer time and the forecast delay of late evidence, making waiting an explicit decision with a closed-form threshold. Evidence is fused only when its location, time and hazard tag are admissible; each decision carries a conformal set and provenance record. We compare it against budget-matched simple rules given the same information, across nine synthetic conditions and a real CrisisMMD workload. On synthetic data a simple rule matched or beat the optimizer everywhere: a fixed rule under bandwidth limits (4.2 versus 4.4 errors); a derived delay threshold cutting 95th-percentile latency from 1,565.0 s to 26.1 s; the admissibility filter, not the optimizer, driving the gain under misaligned reports (9.4 against 20.2). Forged metadata erases that protection; a reused conformal threshold drops coverage from 0.883 to 0.745; measured LTE traces inflate tail latency 8.7×; six controllers cost the exact knapsack 20.0 errors against greedy's 3.9. On real data the ranking reverses — across seven event splits the optimizer leads the fixed rule in 15 of 21 comparisons — while the exact solver and a greedy heuristic on the same utility return identical allocations, and one real modality scoring well on every value-function term carries no signal. The formulation earns its place by naming the quantities and supplying the threshold, not the solver.**

***Index Terms*—Edge AI, degraded connectivity, multimodal fusion, evidence admissibility, conformal prediction, adversarial metadata, real disaster data.**

I. Introduction

A flood, a wildfire or an earthquake produces satellite imagery, field photographs, public reports, weather measurements and infrastructure alarms inside the same hour, and almost none of it arrives together. The image waits for the next satellite pass. The field report describes the neighbouring valley. The network that would carry the raw upload is the same network the event has just damaged. A system that waits for everything can be perfectly accurate and completely useless, because the decision it supports was needed an hour ago.

That is an edge problem before it is a machine learning problem. The interesting question is not which classifier to run but which evidence deserves the bandwidth that remains, whether a late observation is worth waiting for, and whether a given report belongs to this incident at all. Those three decisions are coupled: waiting for an image consumes the time budget that would have let other evidence age gracefully, and admitting a report from the wrong event corrupts the fusion that the budget was spent to enable.

This paper formulates the three decisions jointly and then, more importantly, tries hard to find out whether the formulation earns its complexity. We treat the coupled problem as evidence budgeting across concurrent incident windows and solve it exactly, and we then measure the resulting policy against the simple rules a practitioner would otherwise deploy. Crucially, those rules are given the same information the optimizer receives: the same delay forecast, the same incident metadata, the same byte budget. Much of the apparent value of an adaptive method is really the value of the information it is fed, and the only way to separate the two is to feed a simple rule the same thing.

The answer we obtained is not the one we set out to demonstrate. In every synthetic condition we tested, a simple rule derived from the formulation matched or beat the optimizer that the formulation justifies. On real data the picture changes again: against that same fixed rule the optimizer is ahead in 15 of 21 comparisons across seven event splits, while a single-modality baseline the formulation would never have suggested beats them both on errors, for a reason no term in the formulation can represent. We report that result, identify the mechanism in each case, and draw the consequence: the formulation's value lies in telling a practitioner which quantities matter and supplying the threshold, not in the solver that consumes them.

We also report two failure modes that bear directly on deployment in contested environments. The admissibility rule that keeps foreign reports out of fusion is exactly as strong as the adversary's inability to forge metadata, and its protection degrades smoothly to nothing as that capability grows. And the conformal guarantee that makes the system's uncertainty interpretable does not survive a change of operating condition; reusing a single threshold silently costs coverage precisely when conditions degrade.

The contributions are as follows.

- A formulation of evidence budgeting across concurrent incident windows as a multiple-choice knapsack over a shared byte budget, whose latency term combines transfer time with a forecast of evidence arrival. Waiting becomes an explicit decision with a closed-form delay threshold, and collapsing that threshold to a constant yields a rule deployable without any solver.
- An admissibility rule on location, time and hazard tag applied before evidence is valued or fused, together with a provenance envelope recording every input, version, action and rule behind a decision.
- A validation protocol that compares the method against information-matched simple rules across nine operating conditions, eight of them degraded, with two pre-registered tests, an independently coded reference implementation, and component ablations.
- A reduced version of that comparison on real multimodal disaster data, repeated over seven event splits, which reverses the synthetic verdict against the fixed rule in direction while showing that the outcome in any single condition depends on which disasters are held out, and which falsifies two assumptions made in this paper: the conformal guarantee and the byte model.
- An adversarial evaluation of the admissibility rule under metadata forgery, and a measurement of conformal coverage loss when one calibration threshold is reused across conditions.
- A reproducible artifact in which every table and figure regenerates from frozen result files, with the negative results reported in full.

The system is restricted to analyst decision support. It does not dispatch, order evacuation or allocate aid.

II. Related Work and Positioning

A. Multimodal Disaster Intelligence

xBD supports building-damage assessment from pre- and post-event satellite imagery [1]. FloodNet and Sen1Floods11 support aerial and radar-based flood understanding [2,3]. CrisisMMD provides paired text and images from seven natural disasters [4], and joint text-image representations improve humanitarian-category prediction [5]. These resources define what can be learned from disaster data. None of them defines what should be transmitted, or withheld, when the link degrades.

B. Edge-cloud Collaboration and Communication-efficient Inference

Edge computing moves selected computation toward the data [6,7], collaborative inference partitions it according to device and network state [8], and communication-efficient edge AI treats transmission as a first-order constraint rather than an implementation detail [9]. Recent multimodal systems select what to transmit adaptively [10,11], and in disaster response specifically AVERY adapts the split and compression of a vision-language model on unmanned aerial vehicles to operator intent and fluctuating bandwidth [12]. These methods optimize a single stream or a single device. They do not allocate one budget across concurrent incidents, reason about whether a late observation is worth waiting for, or check whether the evidence belongs to the incident being decided.

C. Uncertainty, Abstention and Trust

Incomplete-modality learning addresses absent inputs [13]. Calibration makes probabilities interpretable [14], selective prediction adds a reject option [15], and split conformal prediction constructs sets with a marginal coverage guarantee under exchangeability [16,17]. That guarantee does not survive distribution shift on its own [18], which is why we calibrate per operating condition and, in Section VI-E, measure what reuse costs.

D. What Is and Is Not Claimed

No individual component here is new. The combination is: one budget shared across incidents, an explicit wait-or-act decision, admissibility checked before valuation, and a status mapping in which severity can raise priority but never confirm a finding. The more unusual feature of this paper is methodological. We evaluate the combination against the alternatives a practitioner would otherwise deploy, give those alternatives the same information, and report the outcome when it goes against the proposed method. Patent literature narrows the novelty claim further: disaster-response systems that structure multimodal on-site data and reason over a disaster-chain knowledge base [19], and edge-cloud schemes combining evidential uncertainty, adaptive modality weights and temporal decay [20], both predate this work. We do not claim to be the first disaster-specific multimodal edge-cloud framework.

III. System Model and Problem Formulation

A. Incident Windows and Admissible Evidence

An incident window i has a grid cell, a decision time and a hazard tag. Each evidence item m for that window carries its own cell, observation time and hazard tag, and is admissible only when all three are compatible with the incident's. Otherwise the only permitted action is to hold it. In the reference configuration the time tolerance is 7,200 s for imagery, 1,800 s for text and 600 s for sensor data, and the cell tolerance is one grid step. The rule reads metadata only and never touches labels. Missing evidence is represented explicitly and is never imputed, because an imputed modality is indistinguishable from an observed one once it reaches fusion.

TABLE I
Notation.

Symbol	Definition
i, s	Incident window; slot of K concurrent windows sharing a budget and a link.
m	Evidence modality: image, text, or sensor.
p_im(y)	Calibrated class probabilities from the representation actually sent.
ŝ_i	Edge estimate of incident severity.
r_im, q_im	Estimated source reliability and observed input quality, in [0, 1].
f_im	Recency factor, exp(−Δt_im / τ_m).
C_im(a), L_im(a)	Serialized byte cost and latency of action a.
D̂_im	Forecast arrival delay of evidence not yet at the edge.
B, R	Per-window byte budget and link rate.

λ_b, λ_ℓ	Byte-cost weight and latency penalty per second.
$\hat{C}(x), \hat{q}$	Conformal set and calibrated threshold at error level α.

B. Actions, Objective, and Why This Is a Knapsack

For each admissible item the policy chooses hold, summary (calibrated local probabilities and metadata), feature (a quantized embedding), or raw transmission. The K windows in a slot share one budget and one link, so the policy maximizes summed evidence value minus normalized byte cost and a latency penalty. The latency term counts transfer time on the shared link and, for an item not yet at the edge, its forecast arrival delay. Holding is not deletion; the item may remain in a local encrypted queue. Because each item contributes exactly one action and the slot shares a single budget, the problem is a multiple-choice knapsack.

Evidence value combines estimated severity, reliability, recency, an action-fidelity multiplier, local uncertainty, input quality and disagreement with the other admissible evidence. The uncertainty term is a proxy for information gain rather than an estimate of it, and we do not claim otherwise. For an image that has not yet arrived no local output exists, so its uncertainty term borrows the uncertainty of the evidence already present, its quality term uses the prior mean, and its disagreement term is zero.

C. The Wait-or-act Decision

Because a late item carries its forecast delay inside the latency term, waiting is not a separate heuristic bolted on beside the allocator. When the slot budget is not binding, the objective selects a late image exactly when some non-hold action has positive net utility, and that condition rearranges into a threshold on the forecast delay. The threshold rises with severity, reliability and the uncertainty of the evidence already in hand, and falls with byte cost and with the latency penalty. The coefficient λ_ℓ therefore states how many seconds of delay one unit of evidence value is worth, and sweeping it traces an accuracy–latency frontier. When the budget binds, the knapsack trades waiting against other windows' needs. A decision that waits also ages the evidence it is holding, which fusion accounts for.

The practical consequence matters more than the derivation. Collapsing the threshold to a constant gives a rule that needs no optimizer at all: for the image-summary action, take the median of the threshold's right-hand side over calibration windows whose image is pending, fix that cutoff before testing, and wait for a late image only when its forecast delay falls below it. Section VI-B compares that rule against the policy it came from.

D. Properties

Proposition 1 (feasibility and exactness). Round each byte cost up to a multiple of $u = KB/2000$ and solve the slot problem by dynamic programming over the rounded budget. Every returned allocation satisfies the true budget, because rounding up can only overstate cost, and the allocation is optimal for the rounded problem. Since each of the 3K items loses at most u bytes of capacity to rounding, the allocation is also optimal for the original problem whenever that problem has an optimum leaving at least 3Ku bytes unused, which is 0.9% of the slot budget at $K = 6$.

Proposition 2 (coverage within a condition). If calibration and test windows are exchangeable within an operating condition, the conformal set contains the true class with probability at least $1 - \alpha$ for that condition. This is the standard split-conformal guarantee [16]. No guarantee is claimed across conditions, and Section VI-E measures what is lost when a single threshold is reused. Section VI-L shows something stronger and less comfortable: on real data the exchangeability this proposition assumes fails within a condition, because calibration and test windows come from different disasters. The carve-out for conditions does not cover that.

Remark (severity cannot confirm). Severity appears only in the rule that routes a case to human review, never in the rule that assigns a confirmed status. A severe but ambiguous case can therefore be escalated but never confirmed. This is a deliberate constraint: severity is the quantity most likely to be correlated with operator anxiety and least likely to be well estimated, and letting it raise confidence would be the easiest way to build a system that looks decisive and is wrong.

IV. Method

Each edge node runs a lightweight model producing class probabilities calibrated by temperature scaling on calibration events [14]. The cloud holds one model per representation: a feature model trained only on the transmitted quantized embedding, and a raw model trained on the full observation. No representation is ever scored with information it did not carry, which is a correction of an error in an earlier version of this work and is described in Section X. Severity is estimated at the edge from the local probabilities of the admissible evidence already present.

Disagreement is the Jensen–Shannon divergence between an item's local probabilities and the mean of the other admissible evidence. For each window the 64 action combinations over three modalities are enumerated with their utilities and byte costs; the dynamic program then processes windows in turn, keeping the best total utility and a back-pointer at every rounded budget level, and the allocation is recovered by backtracking. All-hold is permitted only for a window with no admissible evidence. Per window the policy evaluates 64 combinations and the slot program costs $O(K \cdot 64 \cdot 2000)$ operations, about 0.8 million at $K = 6$.

Selected posteriors are combined with a weighted logarithmic opinion pool over normalized weights. Because the weights are normalized, adding a corroborating modality changes the posterior only through its content and not through the number of sources; corroboration is handled in the status rule instead, where it belongs. If a decision waits for an image, the evidence it holds ages by the waiting time before fusion.

The fused posterior becomes a split-conformal set using the least-ambiguous set-valued classifier score [16], with the threshold calibrated separately for each operating condition on calibration events processed under that condition. When the threshold falls below 0.5 a set can contain at most one class, so a non-singleton set is an empty set: no class reaches the threshold and the method abstains rather than guessing.

TABLE II
Operational statuses. Thresholds are configuration parameters, not learned quantities.

Status	Rule	System behaviour
Human review required	$\hat{s} \geq 0.70$ and (set size $\neq 1$ or conflict)	Escalate with reason codes.
Insufficient evidence	Set size $\neq 1$	Withhold a categorical recommendation; queue to seek evidence.
Conflicting	Maximum pairwise JS divergence ≥ 0.15	Preserve competing evidence; queue.
Confirmed	Singleton set, confidence $\geq$ 0.90, at least two modalities	Place high in the analyst queue; no autonomous dispatch.

Likely	Singleton set otherwise	Monitor; present uncertainty.

Every decision is written to a provenance envelope recording window, event and slot identifiers, the incident's admissibility reference, each item's observed time, cell, hazard tag, availability, admissibility, action, bytes, reason code and a digest of the source observation, the model, policy and calibration versions, the probabilities, the conformal set, the status and the rule that fired, and a digest of the envelope itself. Provenance completeness is computed by checking those fields for every selected item rather than asserted. A digest links a decision to retained source data; it does not establish that the source is true, and nothing in the envelope should be read as doing so.

V. Reference Implementation and Evaluation Protocol

A. Implementation and Integrity Tests

The implementation is a set of deterministic Python modules; the accompanying notebook calls those modules rather than duplicating them. A configuration file fixes seeds, event splits, conditions, the byte model, recency scales, admissibility tolerances, policy coefficients and thresholds. Five integrity properties are tested rather than assumed: the transmission policy is unchanged when test labels and generating severities are permuted; budget-aware methods never exceed a slot budget; the admissibility rule never rejects aligned evidence; the fixed rule with admissibility never sends an inadmissible item; and the derived wait threshold is computed from calibration events only, is unchanged under label permutation, and causes the rule to wait exactly for images whose forecast delay lies below it.

B. Workload and Network Model

A synthetic generator creates three classes and image, text and sensor feature vectors for 12 event identities with 180 windows each, every event carrying a grid location and a hazard tag. Edge models see a prefix of each vector, the raw cloud model sees all of it, and the feature model sees a quantized principal-component embedding. Byte sizes are assumptions: summaries 184–224 bytes, features 768–2,048 bytes, raw images 220–620 kB, raw text 350–1,800 bytes, raw sensor records 220–900 bytes — assumptions that Section VI-L measures against real payloads and finds wrong. Events D00–D05 train the models, D06–D07 calibrate them, and D08–D11 are held out for testing, so no test event contributes to any fitted quantity. The severity estimator's mean training R^2 across seeds was 0.10, which is weak and is treated as such throughout.

Connectivity is modelled, not packet-emulated, and this is the most important limitation of the study to keep in view. Windows form slots of $K = 6$ by decision time within an event. Each slot shares a budget of $K \cdot B$ bytes and one link, served in order of estimated severity, so queueing delay applies to every method including the unconstrained reference. In the delayed-imagery condition 70% of images arrive after a log-normal delay with a 900 s median and the policy sees only a forecast with multiplicative log-normal error ($\sigma = 0.4$). There is no MAC layer, no routing and no contention model in the base configuration; what it captures is the coupling between byte volume, a shared serving order and decision latency. Sections VI-H and VI-I relax two of those assumptions, adding loss with retransmission and capacity replayed from third-party measurements.

C. What the Protocol Asks

The evaluation asks seven questions. Does adaptive selection improve on simple budget-compliant rules when bandwidth is the only constraint? How does the wait-or-act decision trade errors against latency? Does admissibility keep misaligned evidence out of fusion, and what happens when an adversary can forge the metadata it reads? How do calibration and the review queue behave across conditions? And when the shared budget actually binds, does exact optimization beat simple allocation? Sections VI-H to VI-J add a sixth, asked after the others were answered: do these conclusions survive a link that loses packets and varies as a measured one does, and controllers that cannot coordinate? Section VI-L adds a seventh, and the one that bounds all the others: do they hold when the workload is not generated but measured? The second and third questions are each answered against a rule that receives the same information as the optimizer: the delay forecast and the incident metadata respectively.

Each seed provides 720 held-out test windows. We report the mean of five seeds with two-sided 95% t intervals and paired seed-level differences. These intervals describe benchmark variability, not population inference, and with five seeds several of them include zero; we say so wherever it is true rather than only where it is convenient. The questions were revised after an audit of an earlier version of the prototype and were not pre-registered. The information-matched baselines and the review-rule sensitivity analysis were added after a further audit; the wait threshold was fixed on calibration events and the review settings were left unchanged. Two later experiments were pre-registered with frozen protocol files whose SHA-256 digests are checked at run time, and the adversarial sweep in Section VI-D was designed afterwards and is reported as exploratory.

TABLE III
Methods compared. The derived threshold and the fixed rule with admissibility exist specifically to give a simple rule the same information the optimizer receives.

Method	Description	Budget
Image only / text only	Local summary of one modality.	Compliant
Transmit all	Raw evidence for every available modality.	Reference; ignores budget
Confidence gate	Raw when local confidence < 0.72, else summary; downgraded until the window fits.	Compliant
Fixed compression	Feature for every available modality.	Compliant
Fixed rule	Image summary, raw text, raw sensor.	Compliant
Random within budget	Uniform choice among feasible action combinations.	Compliant
Wait if uncertain	Fixed rule, but wait for a late image only if text–sensor confidence < 0.90.	Compliant
Derived threshold	Fixed rule, but wait only if forecast delay < T^*, the median closed-form threshold on calibration events (196–219 s across seeds).	Compliant
Fixed rule + admissibility	Fixed rule with the admissibility filter of Section 3.1.	Compliant
Proposed	Shared-budget knapsack with	Compliant

admissibility and forecast delay.

TABLE IV
OPERATING CONDITIONS.

Condition	Manipulation
Normal	10 Mbps link; 360 kB per-window budget.
Low bandwidth	256 kbps; 4,000-byte per-window budget.
Delayed imagery	2 Mbps; 70% of images late (median 900 s); forecast only.
Intermittent outage	512 kbps; 12 s interruption for 35% of windows.
Image missing / text missing	Remove all images / all text.
Noisy text	Soften text probabilities for 35% of windows; imperfect quality indicator.
Contradictory evidence	Point text toward a wrong class for 35% of windows; imperfect reliability indicator.
Misaligned evidence	Replace 30% of text items with reports from other events; 25% of those carry compatible metadata.

Among the general baselines only the proposed method applies the admissibility rule, so the whole-system comparison under misaligned evidence does not isolate it; the fixed rule with admissibility and the corresponding ablation do. Budget-compliant methods never exceeded a slot budget in any run, and provenance completeness was 100% for every method and condition.

VI. RESULTS

A. When Bandwidth Is the Only Constraint, a Fixed Rule Is Enough

TABLE V
LOW BANDWIDTH (256 KBPS, 4,000 BYTES PER WINDOW). *IGNORES THE BUDGET.

Method	Errors of 720 [95% CI]	Bytes	p95 s	Queue
Image only	51.8 [38.8, 64.8]	224	0.27	5%
Text only	87.4 [52.1, 122.7]	196	0.26	6%
Transmit all*	4.4 [2.3, 6.5]	422,797	84.25	31%
Confidence gate	5.4 [3.5, 7.3]	977	0.45	28%
Fixed compression	4.6 [0.8, 8.4]	3,840	0.94	29%
Fixed rule	4.2 [2.4, 6.0]	1,854	0.59	31%
Random within budget	20.4 [8.8, 32.0]	1,725	0.61	25%
Proposed	4.4 [1.3, 7.5]	3,590	0.92	30%

Among budget-compliant methods the policy and the fixed rule made statistically indistinguishable numbers of errors: a paired difference of +0.2 [-2.8, 3.2]. What the policy did do was spend 1,735 more bytes per window, with a 95% interval of 1,715 to 1,755, and add 0.33 s to tail latency, buying image features that do not change the decision under this byte model. Transmit-all reaches 84.2 s at the tail precisely because six windows share one link. The same pattern holds under normal connectivity, outages, missing modalities, and noisy or contradictory text (Appendix A, Table XVII). When no decision involves waiting or admissibility, the optimizer adds cost rather than value. Sections VI-H and VI-I show that this extra cost is not free once the link is modelled more honestly: under packet loss and under a measured variable link, the bytes the optimizer spends turn into tail latency far faster than the fixed rule's do.

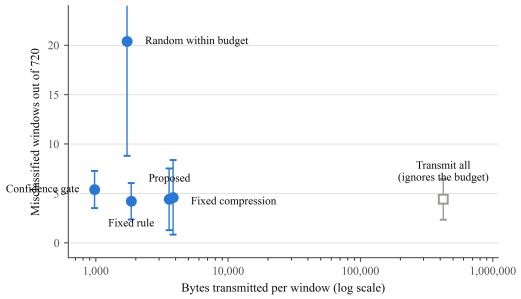


Fig. 1. Low bandwidth: misclassified windows against bytes per window, mean of five seeds with 95% intervals. The hollow marker ignores the budget. Four budget-compliant methods sit within a hair of each other on accuracy while spanning a factor of four in bytes. The two single-modality references are omitted to keep that cluster legible: image-only makes 51.8 errors and text-only 87.4, both in Table V.

B. Waiting for Late Evidence: The Threshold Matters, the Solver Does Not

TABLE VI
DELAYED IMAGERY. *IGNORES THE BUDGET.

Method	Errors of 720 [95% CI]	Median s	p95 s	Image sent
Image only	51.8 [38.8, 64.8]	685.06	1,878.8	100%
Text only	87.4 [52.1, 122.7]	0.15	0.1	0%
Transmit all*	15.2 [10.2, 20.2]	689.84	1,885.0	100%
Confidence gate	18.8 [15.1, 22.5]	685.07	1,878.8	100%
Fixed compression	15.2 [9.3, 21.1]	685.09	1,878.9	100%
Fixed rule	17.4 [12.7, 22.1]	685.07	1,878.9	100%
Random within budget	26.6 [18.6, 34.6]	26.67	1,685.9	68%
Wait if uncertain	12.2 [8.9, 15.5]	74.66	1,565.0	52%
Derived threshold T*	9.4 [6.0, 12.8]	0.18	26.1	31%
Proposed, $\lambda = 0.0004$	9.2 [5.3, 13.1]	0.19	354.6	37%

Methods that always wait paid a median latency near 685.1 s. The policy waited for 37% of images — those with short forecast delays or uncertain partial evidence, exactly as the threshold in Section III-C predicts — and held its median latency at 0.19 s. Against the wait-if-uncertain rule, which ignores the forecast, it changed tail latency by -1,210 s, with a 95% interval of -1,353 to -1,068. Removing the latency penalty made it wait for every image, confirming the term is doing the work.

But the latency reduction does not require the optimizer, and this is the point of the section. The derived-threshold rule sends the fixed rule's image summary whenever the forecast delay falls below one calibration-derived constant. It waited for 31% of images and made 9.4 errors at 26.1 s tail latency. Relative to it the policy changed errors by -0.2 [-2.4, 2.0], tail latency by +328.5 s [252.3, 404.6] and bytes by +683 [643, 722]. The policy's longer tail comes from per-window adaptivity: it waited more than 300 s for 5.9% of windows against 0.5% for the constant rule, when uncertain partial evidence raised the value of a late image. In this benchmark those long waits bought nothing.

The accuracy comparison needs a caveat that is easy to miss and changes the conclusion. The policy changed errors by -8.2 [-11.8, -4.6] against the fixed rule and by -3.0 [-5.3, -0.7] against wait-if-uncertain. Those gains exist because the benchmark decays the weight of text and sensor evidence while a decision waits. With that decay removed, the always-wait baselines return to normal accuracy — 4.4 errors for the fixed rule, 4.4 for transmit-all and 4.8 for wait-if-uncertain, against 8.6 for the policy and 9.6 for the derived threshold, all read from the sensitivity rows of ablation_results.csv. Without staleness, both forecast-using rules trade four to five errors for their latency reduction. Whether acting early also improves accuracy therefore depends entirely on how fast real evidence goes stale, which this benchmark assumes rather than measures, and which a deployment would have to establish from its own data.

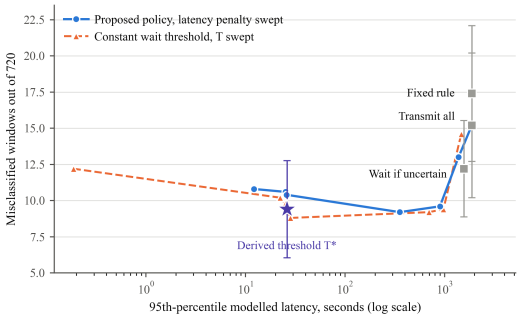


Fig. 2. Delayed imagery: errors against 95th-percentile modelled latency. The solid line traces the policy across latency-penalty values; the dashed line traces the constant-threshold rule across T. The constant-threshold frontier lies on or slightly below the policy's throughout. The star marks T* derived from calibration events, not selected on test data.

C. Admissibility, Not Optimization, Keeps Foreign Reports Out

When 30% of text items came from other events, methods without the admissibility rule fused all of them. The proposed method fused 24% of misaligned items — exactly those whose metadata had been made compatible — and rejected no aligned item. The complete system made 8.8 errors against 20.2 for the fixed rule without admissibility, a paired difference of -11.4 [-21.7, -1.1]. The fixed rule with the same filter made 9.4 errors; relative to it the proposed method changed errors by -0.6 [-3.8, 2.6]. The filter accounts for the gain. Within the proposed system, disabling admissibility added +8.4 errors [-0.5, 17.3], an interval that includes zero, so even the filter's own effect is imprecisely estimated at five seeds.

This test also favours the filter by construction, and saying so is not a formality. Aligned evidence shares the window's cell and hazard tag exactly and is at most 60 s old, far inside the tolerances, while misaligned reports that were not spoofed carry the coordinates of another event on a 400 × 400 grid. Zero false rejections therefore follow from the generator, not from the rule's robustness. Real metadata carries positional error, clock skew and frequently missing geotags, none of which is represented here. The next section addresses the other half of the problem: what the rule is worth when the metadata is not merely noisy but forged.

D. Adversarial Metadata: The Filter Is Exactly as Strong as Forgery Is Hard

The admissibility rule reads metadata and nothing else, so an adversary who can attach plausible location, time and hazard values to a report defeats it by construction. Section VI-C stated that; this section measures it. We hold the misaligned-evidence condition fixed and vary one quantity: the fraction of misaligned reports whose metadata is set to the target incident's own values, which is what an adversary capable of forging metadata supplies. Everything else — the 30% misalignment rate, the link, the budget, the models — is unchanged. This experiment was designed after the earlier results and is exploratory; we report it as a robustness curve, not as a hypothesis test.

TABLE VII
ERRORS OUT OF 720 AS THE SHARE OF MISALIGNED REPORTS CARRYING FORGED METADATA RISES. MEANS OF FIVE SEEDS.

Method	**0%**	**25%**	**50%**	**75%**	**100%**
Fixed rule (no filter)	20.2	20.2	20.2	20.2	20.2
Fixed rule + admissibility	7.0	9.4	11.4	15.0	20.2
Proposed	6.4	8.8	10.2	12.8	17.2

The result is a clean, almost linear collapse. With no forgery the filter is worth a great deal: 7.0 errors against 20.2 for the unfiltered rule. With complete forgery it is worth nothing at all: 20.2 errors against the unfiltered rule's 20.2, a difference of +0.0. In between, protection falls away in proportion to the adversary's capability. The admitted fraction tracks the forged fraction almost exactly — 24.1%, 48.4% and 72.4% against 25%, 50% and 75% — which says the rule admits the forged items and nothing else, and rejects no aligned item at any point on the sweep. The optimizer again contributes nothing to robustness: the proposed method tracks the filtered fixed rule within 0.6 to 3.0 errors across the whole range.□

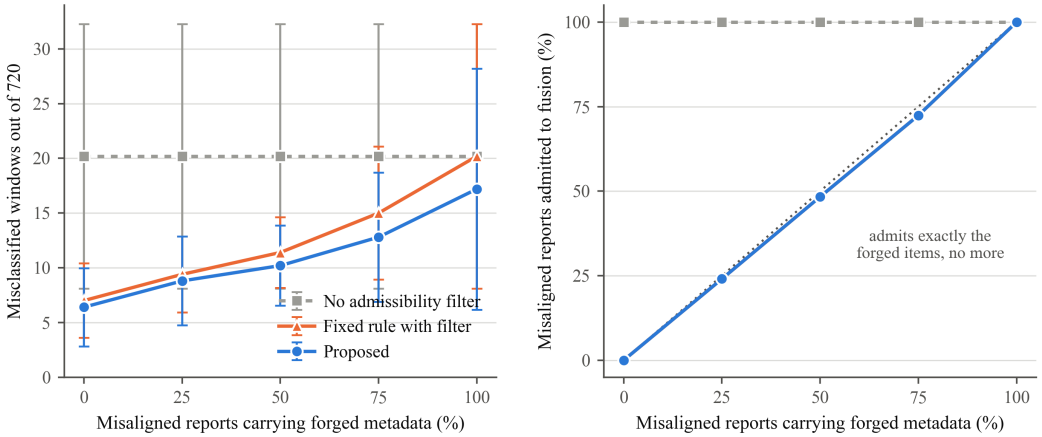


Fig. 3. Adversarial metadata forgery. Left: errors against the share of misaligned reports carrying forged metadata, with 95% intervals. Right: the share of misaligned reports admitted to fusion; the dotted line is the identity. The two filtered methods coincide exactly on the right panel, so the orange series is hidden beneath the blue.□

Two practical consequences follow. First, there is no safe operating point: the degradation has no threshold below which the filter can be relied upon, so a deployment cannot argue that a low expected forgery rate makes the rule sufficient. Second, metadata admissibility is a cheap and effective filter against accident — misrouted reports, stale caches, misconfigured clients — and is not a security control. Where an adversary is in the threat model, admissibility must be backed by something that forgery cannot cheaply defeat, such as signed observations or attested sources, and the cost of that machinery belongs in the byte budget this paper is about. We have not evaluated such a scheme, and the sweep models metadata substitution only: not content-level attacks, coordinated multi-item campaigns, compromised edge nodes, or attacks on the models themselves.

Metadata forgery is one adversary capability. Controlling what a source says is another, and the two need not be held by the same attacker. We cross them: the forgery rate of Section VI-D against a compromised text source that points 35% of reports toward a wrong class, which the benchmark already models as the contradictory-evidence condition. The grid includes the reference forgery rate of 25%, where it reproduces Section VI-C exactly, so the two tables can be read against each other.□

TABLE VIII

ERRORS OUT OF 720 UNDER TWO ADVERSARY CAPABILITIES. THE FIRST FOUR NUMERIC COLUMNS HAVE AN UNCOMPROMISED TEXT SOURCE; THE LAST FOUR HAVE A COMPROMISED ONE CORRUPTING 35% OF REPORTS. WITHIN EACH GROUP THE FORGERY RATE RISES FROM 0% TO 100%, AND THE 25% COLUMN IS THE REFERENCE CONDITION OF SECTION VI-C, WHICH IT REPRODUCES EXACTLY.

Method	0%	25%	50%	100%	0%	25%	50%	100%
Fixed rule (no filter)	20.2	20.2	20.2	20.2	19.8	19.8	19.8	19.8
Fixed rule + admissibility	7.0	9.4	11.4	20.2	10.6	14.0	16.0	19.8
Proposed	6.4	8.8	10.2	17.2	10.2	12.8	14.8	18.4

□

Forgery alone is sufficient. With an uncompromised source, full forgery takes the filtered rule from 7.0 errors to 20.2, which is exactly the unfiltered rule's 20.2. Adding a compromised source on top does not compound the damage: with both capabilities the filtered rule reaches 19.8, no worse than forgery alone. The filter's value is bounded by the adversary's forging capability and by nothing else in this benchmark. A compromised source is damaging on its own — it takes the filtered rule from 7.0 to 10.6 errors with no forgery at all — but it is a different problem, and admissibility was never the defence against it.

E. Conformal Coverage Does Not Survive a Change of Condition

Calibrating per condition kept empirical coverage between 0.883 and 0.898 against a nominal 0.90; the shortfall is consistent with having only two calibration events. Reusing the normal-condition threshold is where the trouble starts. Coverage falls to 0.794 under contradictory evidence and to 0.745 with images missing — a loss of 13.8 percentage points in the condition where an operator is most likely to be relying on the interval. Proposition 2 should not be stretched across conditions, and a deployment that calibrates once at commissioning and never again will be quietly miscalibrated exactly when the network degrades. Section VI-L shows that calibrating per condition is necessary but not sufficient: on real data the guarantee also fails across events, which per-condition recalibration cannot repair.

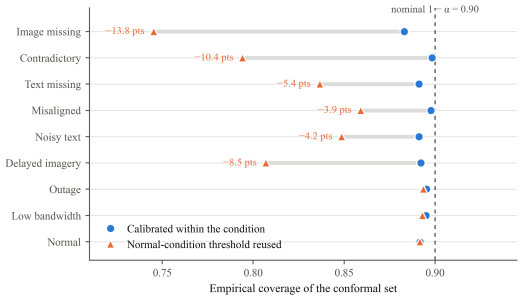


Fig. 4. Empirical coverage of the conformal set, by condition, calibrated within the condition against the normal-condition threshold reused. Conditions are ordered by the size of the loss. The guarantee holds within a condition and does not transfer across them.

F. The Review Queue Is the Real Operational Cost

TABLE IX

REVIEW-RULE SENSITIVITY AT $\alpha = 0.1$, ALL NINE CONDITIONS. THE CONFLICT RULE ROUGHLY TRIPLES THE QUEUE AND CATCHES ALMOST NOTHING EXTRA.

Condition	Errors	Queue	Caught	Queue	Caught

		(JS ≥ 0.15)		(conflict off)	
Normal	4.6	30%	4.6	11%	4.6
Low bandwidth	4.4	30%	4.4	10%	4.4
Delayed imagery	9.2	25%	9.0	11%	7.8
Image missing	12.2	23%	11.2	11%	10.0
Text missing	12.4	22%	12.0	11%	10.0
Noisy text	6.0	33%	6.0	11%	6.0
Intermittent outage	4.2	30%	4.2	10%	4.2
Contradictory	12.4	53%	12.2	10%	11.2
Misaligned	8.8	30%	8.6	10%	8.2

About 30% of windows entered the combined abstention and review queue under low bandwidth, roughly 61 queued windows per caught error, and 53% under contradictory evidence. Under low bandwidth 98% of queued windows were correctly classified, meaning almost all of that reviewer effort confirmed a decision the system had already got right. Conformal abstention alone queued about 10% of windows and still caught every error; the prespecified conflict rule nearly tripled the queue without catching an additional one. Across conditions the conflict rule added 11 to 43 percentage points of queue load for between zero and two additional caught errors per 720 windows. We keep the prespecified settings in the main results rather than tuning after the fact, but the implication for deployment is plain: the conflict threshold is a staffing decision and should be set from reviewer capacity, not from a default. The severity-based escalation rule was nearly inert, firing on 0.07% of windows against the 28% the generating severity value would have escalated, because the severity estimator is weak. Its first clause alone, ignoring the conjunction, reaches only 0.22%, so it is the estimate and not the rule shape that makes the escalation rare.

G. When the Shared Budget Binds

None of the conditions above makes the shared budget bind, which is the only regime in which the knapsack can differ from a per-window rule. At the reference difficulty it could not have mattered: on calibration events, sending only summaries erred on 0.56% of windows against 0.44% for the unconstrained fixed rule. So we froze a protocol before running the proposed method on a harder condition, with the protocol file and its SHA-256 digest inside the package. Observation noise was scaled by 1.75, the smallest value in a pre-specified grid at which summaries alone err on at least 8% of calibration windows. Models were retrained at that difficulty, the link runs at 256 kbps with K = 6, and per-window budgets range from 700 to 1,900 bytes. Ten seeds. The pre-specified competitor is a shared-budget greedy rule that starts from summaries and upgrades items across the slot in order of local uncertainty; a greedy heuristic on the proposed utility isolates the exact solver. A difference counts only if its paired t-test survives Holm adjustment across the five budgets at 0.05.

TABLE X
CONTESTED SHARED BUDGET (NOISE SCALE 1.75, TEN SEEDS): ERRORS OUT OF 720 BY PER-WINDOW BYTE BUDGET B. *IGNORES THE BUDGET; SHOWN AS A CEILING.

Method	700	900	1,200	1,500	1,900
Summaries only	78.6	78.7	78.6	78.6	78.8
Fixed rule, per window	78.6	75.1	68.9	65.6	64.0
Confidence gate	78.6	76.0	70.0	66.3	64.7
Random within budget	140.4	141.3	127.1	121.1	109.2
Greedy by uncertainty	76.0	71.2	68.9	63.3	61.9
Greedy, proposed utility	96.4	72.3	64.6	60.7	57.2
Proposed (exact knapsack)	96.8	72.6	64.9	59.7	57.1
Fixed rule, ignores budget*	57.6	57.5	57.7	57.5	57.4

The optimizer did not beat the pre-specified competitor, the greedy uncertainty ranking, at any budget. At 700 bytes it was worse by +20.8 errors [15.4, 26.2] with Holm $p < 0.001$. At the three largest budgets it led by -4.0, -3.6 and -4.8 errors, and none of those survived adjustment (Holm p 0.075 to 0.219). The exact solver matched a greedy solver on the same utility to within one error at every budget. One of those five differences, at B = 1,500 bytes, did survive Holm adjustment in the solver's favour ($p = 0.042$), and the results file records it as such. It is 1.0 error in 720, or 0.14% of windows, and we do not regard that as a benefit worth an exact solver — but the decision rule set out above was met once, and it would be wrong to say the solver never beat greedy.

The tight-budget failure has a clean mechanism, and it is the one robust lesson of this experiment. At 700 bytes the optimizer sent nothing for the image in 27% of windows and nothing for the text in 30%, and held at least one modality in 47%, in order to buy raw sensor records elsewhere in the slot. It did that because the value function prices a held modality at zero rather than at the cost of losing it. Exact optimization does not repair a misspecified objective; it exploits it more efficiently than a greedy rule can.

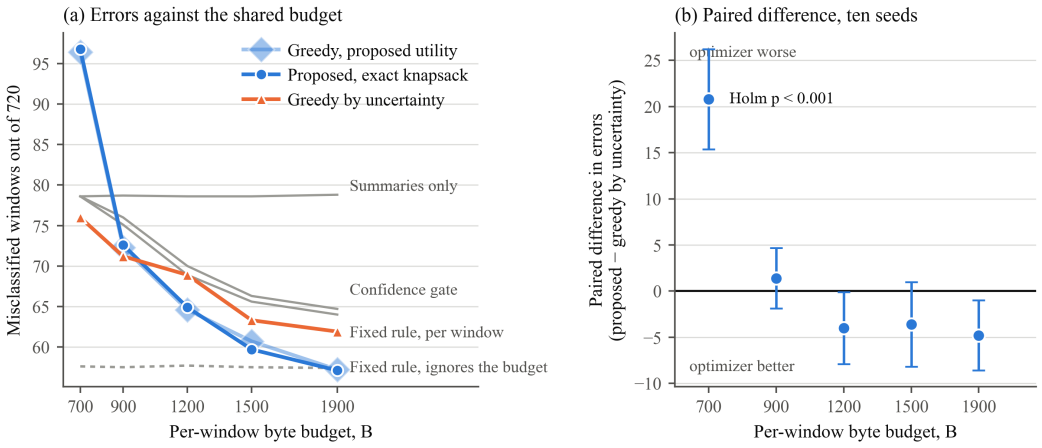


Fig. 5. Contested shared budget. (a) Errors against the per-window budget; marginal intervals are omitted because between-seed variation dominates them. The

pale diamonds are the greedy heuristic on the proposed utility, which tracks the exact knapsack closely enough to be read as a halo around it. Random-within-budget is omitted from the panel because it runs from 140.4 to 109.2 errors and would flatten every series compared here; it is in Table X. (b) The paired difference the analysis actually tests, with 95% intervals over ten seeds. Only the tight-budget failure survives Holm adjustment.□

An exploratory variant designed after these results adds a summary floor: every eligible item is sent at least as a summary and the knapsack chooses only upgrades. We froze a second protocol and tested it on ten new seeds. The floor removed the tight-budget failure. Its advantage over greedy-by-uncertainty replicated in direction at the three largest budgets (-3.0, -3.1 and -3.4 errors) but did not survive adjustment (Holm p 0.087 to 0.353), and was absent at the two tightest budgets (+0.2 and -0.9). The exact solver again matched a greedy solver with the same floor to within one error. The proposed value function may carry a benefit of roughly 0.4% of windows at moderate budgets; this benchmark does not establish it, and the exact solver contributes nothing measurable.

H. Packet Loss Makes the Optimizer's Extra Bytes Expensive

The results so far treat the link as a constant-rate pipe, which is the assumption most evaluations of this kind make and the one this journal's readership is most likely to question. Two experiments relax it. Both are exploratory, designed after the main results, and both leave every earlier number untouched. Each variant inherits the perturbation stream of the condition it modifies, so it is a controlled comparison rather than a differently drawn workload, and setting the loss rate to zero with no trace reproduces the preceding sections exactly. We check that rather than assert it: at zero loss every method's errors, bytes and latencies equal the normal-condition values of Table XVII to every printed digit.

The first adds packet loss. A payload is split into 1,500-byte packets, each lost independently with probability p and retransmitted, so the bytes actually placed on the wire grow by $1/(1 - p)$ and each lost packet additionally costs one 200 ms retransmission timeout. The expected number of lost packets grows with the payload, so the time penalty is super-linear in bytes. That is precisely the regime in which sending more than you need stops being merely wasteful and starts being slow.

TABLE XI
95TH-PERCENTILE MODELLED LATENCY IN SECONDS AGAINST PACKET LOSS, AT THE NORMAL PER-WINDOW BUDGET. *IGNORES THE BUDGET.

Method	0%	1%	2%	5%	10%
Fixed rule	0.09	0.10	0.10	0.12	0.14
Confidence gate	0.33	0.39	0.40	0.41	0.44
Proposed	0.57	0.83	1.23	2.47	4.70
Fixed compression	0.10	0.11	0.12	0.14	0.17
Transmit all*	2.24	2.89	3.64	6.12	10.72

Accuracy is unmoved: every method's error count is the same at 10% loss as at zero, because nothing in this condition ties latency back to the decision. Tail latency is not unmoved, and it separates the methods sharply. The fixed rule goes from 0.09 s to 0.14 s, a factor of 1.5. The proposed policy goes from 0.57 s to 4.70 s, a factor of 8.3. The gap between them widens from 6× to 34×. The optimizer's surplus bytes were close to free under an idealised link; under a lossy one they are the dominant term. This is a cross-layer effect in the ordinary sense: a link-layer parameter changes which application-layer policy a practitioner should deploy, and it changes it against the more sophisticated one.

I. A Measured Link Is Not a Constant One

The second experiment replays measured cellular capacity traces in place of the constant rate [21]. Each trace records the milliseconds at which one MTU packet could be delivered on a real LTE uplink; we replay them unmodified, varying only the starting offset by slot. Comparing a replayed trace against a constant link at the same mean capacity isolates the effect of variability from the effect of speed, which a comparison against an arbitrary constant rate cannot do.

TABLE XII
95TH-PERCENTILE MODELLED LATENCY IN SECONDS UNDER A REPLAYED MEASURED TRACE AGAINST A CONSTANT LINK OF THE SAME MEAN CAPACITY, EVERY METHOD AND EVERY TRACE. *IGNORES THE BUDGET.

Trace	Method	Constant at same mean	Replayed	Inflation
Verizon-LTE-short	Fixed rule	0.24	0.37	×1.5
	Confidence gate	0.23	0.32	×1.3
	Fixed compression	0.26	0.42	×1.6
	Proposed	0.26	0.42	×1.6
	Transmit all*	3.84	7.22	×1.9
Verizon-LTE-driving	Fixed rule	0.24	1.49	×6.2
	Confidence gate	0.23	0.94	×4.0
	Fixed compression	0.25	2.20	×8.7
	Proposed	0.25	2.17	×8.6
	Transmit all*	3.67	20.42	×5.6
TMobile-LTE-driving	Fixed rule	0.23	0.25	×1.1
	Confidence gate	0.23	0.24	×1.1
	Fixed compression	0.23	0.27	×1.1
	Proposed	0.23	0.26	×1.1
	Transmit all*	1.34	3.74	×2.8

Holding mean capacity fixed, real variability inflates tail latency by up to 8.7× across the methods and traces of Table XII. On the Verizon driving trace, whose fifth-percentile per-second throughput falls to 0.25 Mbps, the proposed policy's tail rises from 0.25 s under a constant link of the same mean to 2.17 s when the trace is replayed, and the fixed rule's from 0.24 s to 1.49 s. The short Verizon trace and the faster, steadier T-Mobile trace inflate far less, which is the point: the penalty tracks the depth of the fades, not the average rate. A constant-rate model is therefore optimistic in exactly the conditions this paper is about, and the size of the error depends on a property of the link that a mean does not capture. Ranking the five methods by tail latency gives the

same order under the replayed trace as under its mean-matched constant, for all three traces, so the paper's conclusion survives the more realistic link; what does not survive is the idea that the optimizer's extra bytes are cheap.□

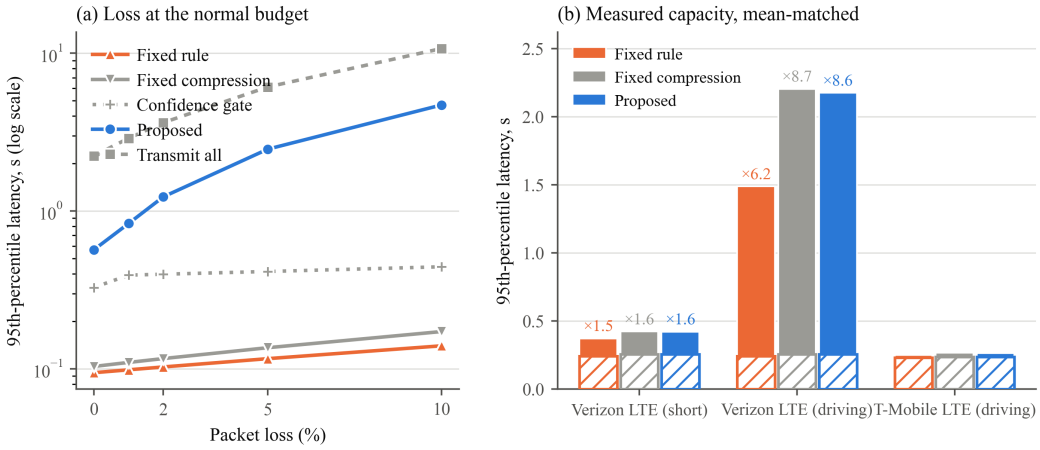


Fig. 6. Link realism. (a) Tail latency against packet loss at the normal budget, every method in the sweep, log scale: the penalty grows with payload, so the policy that sends more is hurt most. (b) Replayed measured capacity against a constant link of the same mean, for the largest inflation, the proposed policy and the fixed rule it is measured against; the other two methods are in Table XII. Hatched outlines are the constant, solid fills the replay, and the annotation gives the inflation factor.□

J. Coordination Is Worth More to the Optimizer Than to the Simple Rule

Every result so far grants the shared-budget formulation its most favourable assumption: one controller owns all six windows in a slot and allocates the whole budget. Field deployments are unlikely to satisfy it. An incident area is more plausibly served by several edge nodes that cannot coordinate per slot over the same degraded link they are trying to conserve. This section removes coordination and measures what it was worth.

The slot's windows are dealt round-robin to C independent controllers, each allocating only its own windows against a share of the byte budget proportional to the windows it owns. The total budget is unchanged and all controllers still share the physical link: partitioning control does not partition the medium. C = 1 is the centralised design and reproduces the earlier results exactly. The contested-budget condition is used because it is the only regime in which the shared budget binds, and therefore the only one in which coordination can matter. As in Section VI-H the variant inherits the contested-budget perturbation stream, and C = 1 reproduces Table X exactly for every method at every budget, which is the check that makes the rest of the table a measurement of coordination rather than of a different draw.

TABLE XIII
ERRORS OUT OF 720 AT B = 900 BYTES AS COORDINATION IS REMOVED. C IS THE NUMBER OF INDEPENDENT CONTROLLERS SHARING ONE SLOT. TEN SEEDS.

Method	C = 1	C = 2	C = 3	C = 6
Summaries only	78.7	78.7	78.7	78.7
Fixed rule, per window	75.1	75.1	75.1	75.1
Greedy by uncertainty	71.2	71.4	72.0	75.1
Proposed (exact knapsack)	72.6	78.3	81.5	92.6

Two per-window methods act as a control. Summaries-only and the per-window fixed rule allocate nothing across windows by construction, so C cannot affect them, and it does not: they move by less than a window across the whole range. That the partitioning leaves them alone is the evidence that it is doing what it claims.

The two coordinated methods separate sharply. Removing coordination entirely costs the exact knapsack +20.0 errors [14.2, 25.8] at B = 900, and the greedy uncertainty ranking only +3.9 [0.8, 7.0]. The penalty is not monotone in the budget: it peaks at B = 900 (+20.0 errors) rather than at the tightest budget B = 700 (+10.2), because at 700 bytes the centralised optimizer already fails badly enough (96.8 errors) that there is little left for decentralization to take away. The consequence is blunt: at C = 6 the proposed method makes 92.6 errors against the greedy rule's 75.1, so the ordering that was roughly a tie under centralisation becomes a clear loss once the centralisation assumption is dropped.

This is the sharpest form of the paper's thesis. The exact optimizer is not merely no better than a greedy ranking; it is more dependent on an architectural assumption that field deployments are least likely to satisfy. A method whose advantage requires per-slot consensus over the very link it is conserving has to justify that consensus as part of its cost, and this benchmark charges nothing for it. A greedy uncertainty ranking degrades gracefully because each controller can apply it to whatever it owns; the knapsack degrades because its value comes from trading bytes between windows that separate controllers no longer see.□

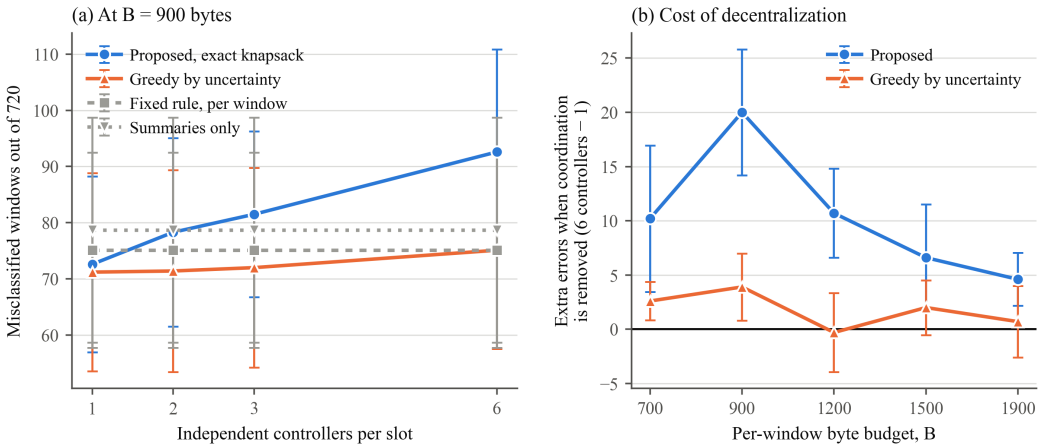


Fig. 7. Removing coordination. (a) Errors against the number of independent controllers at B = 900 bytes, all four methods, with 95% intervals; summaries-only and the per-window fixed rule are flat by construction and serve as controls. (b) The paired cost of full decentralization across the budget grid, shown for the two methods that allocate across a slot and can therefore lose something when the slot is partitioned; for the two per-window methods that cost is identically zero.□

K. What the Policy Costs to Compute

Complexity was stated analytically in Section IV and never measured. It is measured here, on the contested-budget views the benchmark actually produces, timing the allocation step for each method as best-of-five wall clock and recording peak allocation. This is a compute measurement on one machine in one interpreter. It is not an energy measurement, and no power figure is inferred from it.

TABLE XIV

MEASURED EDGE-SIDE DECISION COST AT B = 1,200 BYTES. EACH SEED IS TIMED FIVE TIMES AND THE FASTEST RUN IS KEPT, TO SUPPRESS SCHEDULER NOISE; THE TABLE REPORTS THE MEAN OF THE FIVE SEEDS.

Method	Per slot (ms)	Per window (ms)	Peak allocation (KiB)
Fixed rule	0.0012	0.0002	76
Fixed rule, per window	0.2070	0.0345	76
Greedy by uncertainty	0.4949	0.0825	180
Greedy, proposed utility	15.4829	2.5805	1,329
Proposed (exact knapsack)	7.4428	1.2405	1,329

The gap is larger than any accuracy difference reported in this paper. The fixed rule decides a slot in 0.0012 ms; the exact knapsack takes 7.44 ms, about 6,190 times longer, and holds 18 times the peak memory. On the benchmark's own evidence the optimizer buys no accuracy over the fixed rule when bandwidth is the only constraint, so this is cost without return; on a constrained edge node it is also the cost most likely to matter.

Solve time grows close to linearly in the number of windows sharing a slot, which is what the $O(K \cdot 64 \cdot 2000)$ bound predicts — 1: 0.10 ms, 2: 0.42 ms, 3: 0.88 ms, 4: 1.44 ms, 6: 2.37 ms, 8: 3.24 ms, 12: 4.83 ms — so the analytic claim survives measurement. One result is an implementation artefact rather than a complexity finding and should be read as such: the greedy heuristic on the proposed utility is slower than the exact dynamic program (15.5 ms against 7.4 ms) because the greedy search is a Python loop while the dynamic program is vectorised. A production implementation of either would differ.

L. A Reduced Version of the Comparison on Real Disaster Data

Every result above rests on a generated workload; the only measured input so far is the capacity traces of Section VI-I. This section removes that qualification for the part of the study it can, and the answer is more complicated than the synthetic benchmark suggested. We replace the synthetic workload with one built from CrisisMMD [4]: 3,526 labelled report–image pairs from 7 disasters in 2017, with real text, real photographs, real hazard types, real arrival times recovered from the message identifiers, real class imbalance and real payload sizes. The policies are unchanged and the split is by event. This experiment was designed after the results above, was not pre-registered, and is reported as exploratory.

Both modalities are represented by CLIP embeddings [23], reduced by principal components fitted on training events only — eight components for text and ten for images, matching the dimensions the policy expects. Their own classifiers, measured on the 968 held-out records before any budgeting decision is taken, separate cleanly: the image representation reaches 0.566 accuracy against a majority baseline of 0.490, while the text representation returns 0.490, which is the majority baseline to four decimal places. Text predicts the majority class for every held-out record. An earlier version of this workload used TF–IDF text features and pixel statistics and failed for both modalities, leaving image at the baseline too; replacing it lifted the image side and moved the text side not at all. That is why the text result below is read as a property of the task rather than of the encoder — it survived the encoder upgrade that fixed the other modality. These figures are produced by representation_diagnostics.py and are the raw per-record accuracies, not the per-window error counts of Table XV, which are decided under a budget.

The comparison is reduced, not repeated, and the reductions are forced by the data. CrisisMMD has no sensor stream, so the workload has two modalities rather than three and the per-window action space shrinks from 64 combinations to 16; a fixed-seed placeholder keeps the shared code path running and the runner asserts it is never available. The annotations carry no geolocation, so the spatial term of admissibility reduces to event identity. No severity field exists that is independent of the damage label, so severity is held constant rather than derived from it, which would reintroduce the leakage an earlier version of this prototype was corrected for; the severity term of the value function is therefore inert here. Imagery is never missing in this corpus, so the incomplete-modality machinery is not exercised. Three conditions carry over, not nine: delayed imagery is absent, so the derived constant wait threshold — the paper's clearest positive result — is not tested on real data at all. Three methods are absent for the same reason, including the fixed rule with admissibility, so the misaligned condition here cannot isolate the filter the way Section VI-C does.

Five seeds vary model fitting only. The data, the split and the 968 test windows are identical across them, and the test set is two

disasters, of which one supplies most of it: 795 of the 968 test windows come from Hurricane Irma, and 173 from the Iraq–Iran earthquake. Read what follows as a result about those two events, weighted heavily toward the first. Intervals in this section therefore bound fitting jitter, not sampling error, and several are zero-width because the method concerned is deterministic given the data. They should not be read as the synthetic intervals are read.□

TABLE XV
ALL THREE REAL-DATA CONDITIONS. ERRORS OUT OF 968; THE MAJORITY-CLASS BASELINE IS 494. BYTES AND TAIL LATENCY ARE FOR LOW BANDWIDTH. *IGNORES THE BUDGET.

Method	Normal	Low bw.	Misaligned	Bytes/window	p95 s
Image only	420.0	420.0	420.0	224	0.27
Transmit all*	435.8	435.6	447.4	129,728	30.71
Fixed rule	464.2	465.4	473.4	342	0.29
Confidence gate	437.0	465.8	474.4	1,390	0.56
Random within budget	462.0	465.8	474.6	1,212	0.52
Greedy by uncertainty	466.8	466.8	476.0	2,166	0.63
Greedy, proposed utility	447.0	466.8	462.2	1,896	0.63
Proposed (exact knapsack)	447.0	466.8	462.2	1,896	0.63
Fixed compression	483.0	483.0	486.0	3,072	0.80
Text only	494.0	494.0	494.0	196	0.26

□

Against the fixed rule — the simple rule this paper has used as its comparator throughout — the ranking does not carry over. The optimizer wins under normal connectivity by 17.2 errors [15.8, 18.6] and under misaligned evidence by 11.2 [7.8, 14.6], and loses only under low bandwidth, by 1.4 [0.7, 2.1], which is 0.14% of the test set. That is the reverse of the synthetic finding, and reporting only the low-bandwidth column would have reproduced the synthetic conclusion by selection rather than by evidence.

One assignment of seven events to three splits produced that pattern, and the test set is dominated by one hurricane, so the obvious question is whether the signs belong to the data or to the split. We rotated the event list through 7 pre-specified assignments, each keeping the same 3/2/2 shape, so that every event serves in the test set at least once. The rotations were fixed before running and none was discarded afterwards. Rotation 0 is the split reported above and reproduces Table XV exactly, which is the check that the rest of the table measures the split and not a different pipeline.

TABLE XVI
PROPOSED MINUS FIXED RULE IN ERRORS, BY EVENT SPLIT. NEGATIVE MEANS THE OPTIMIZER WINS. ROTATION 0 IS THE SPLIT OF TABLE XV. COUNTS DIFFER BECAUSE THE EVENTS DIFFER IN SIZE.

Rotation	Test events	n	Normal	Low bw.	Misaligned
0	Irma + Iraq–Iran	968	-17.2	+1.4	-11.2
1	Harvey + Iraq–Iran	1,065	-3.0	-5.4	+2.8
2	Harvey + Maria	1,754	-0.2	+1.0	+1.6
3	California + Maria	1,393	-18.8	-12.8	+8.2
4	California + Mexico	709	-3.0	-1.0	+18.8
5	Mexico + Sri Lanka	273	-7.6	-8.6	-6.4
6	Irma + Sri Lanka	890	-57.2	-33.2	-47.6

The direction survives the rotation and the per-condition pattern does not. The optimizer is ahead of the fixed rule in 15 of the 21 comparisons, and under normal connectivity it wins in every one of the 7. But the two other signs reported above are minority outcomes: the loss under low bandwidth occurs in 2 rotations of 7, and the win under misaligned evidence in 3. What generalises from this corpus is therefore narrower than Table XV alone would suggest: on real data the optimizer is ahead of the fixed rule more often than not, decisively so when the link is not the constraint, and which way any single condition falls depends on which disasters land in the test set. The synthetic verdict is still reversed — there the fixed rule matched or beat the optimizer in all nine conditions — but the reversal is a statement about the direction, not about any particular condition.

The comparison that matters most for the argument of this paper is between the exact solver and a greedy heuristic on the same utility, because that is what separates the formulation from the optimizer it justifies. On real data the two are not close; they are identical. A greedy multiple-choice knapsack over the proposed utility produced the same errors, the same bytes per window and the same 95th-percentile latency as the exact dynamic program in all three conditions and at every one of the five seeds, a paired difference of exactly 0.0 on all three metrics with a zero-width interval. The exact solver is not merely unrewarded here; on this workload it selects the same allocation the heuristic does. A greedy ranking by local uncertainty, which does not use the utility at all, is a different matter: it makes 466.8 errors under normal connectivity against 447.0, so the utility is doing work even though the solver is not.

What does beat everything on errors is image-only, at 420.0 in every condition, ahead of the optimizer by 46.8 errors [46.2, 47.4] under low bandwidth while sending 224 bytes. That result needs two caveats before it is used for anything. Its value is identical in all three conditions and across all five seeds, because the conditions perturb text and bandwidth and image-only consumes neither; it cannot vary, so its constancy is a property of the design and not evidence of robustness. And on the synthetic benchmark the same method makes 51.8 errors out of 720 against the fixed rule's 4.2, which is among the worst in the study. A method that is near-worst on one workload and best on the other is telling us about the workloads, not about itself.□

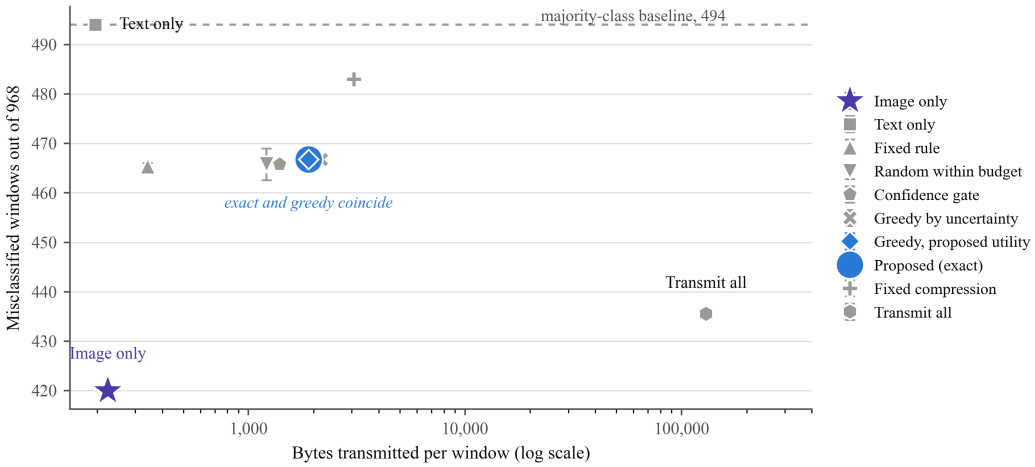


Fig. 8. Errors against bytes on real data under low bandwidth, every method in the comparison, identified by the legend. Intervals are fitting jitter across five seeds on one fixed test set, and are zero-width for the deterministic methods. The exact knapsack and the greedy heuristic over the same utility occupy the same point.□

The reason image-only wins is visible in the same table. Text-only scores exactly 494.0 errors, the majority-class count, in every condition: its classifier predicts one class for all 968 windows. Everything that fuses text pays for it — image-only makes 420.0 errors and the fixed rule, which adds raw text, makes 465.4. On this corpus the text modality carries no signal for the label, and admitting it costs 45.4 errors.

How far that generalises is exactly what we cannot establish, and we would rather say so than imply otherwise. The label is damage annotated from the photograph, so a label defined on one modality failing to be predicted by the other is unsurprising and is narrower than the claim that real corpora contain uninformative sources. The text pathway is also an eight-component reduction of a general-purpose embedding, and we did not vary the encoder or the dimensionality, so we cannot separate a property of the task from a property of this representation. What the data does support is narrower and still useful: when a modality carries no signal, nothing in the value function detects it. Severity, reliability, recency, fidelity, uncertainty, quality and disagreement are all properties of the source or the signal; none asks whether this modality has ever predicted this label. A held-out estimate of predictive contribution is a different kind of quantity and is the one that would have been needed.

Transmit-all is also ahead of the optimizer on errors here, by 31.2 errors [30.2, 32.2], and that comparison is worth stating carefully because it is not free: it sends 129,728 bytes against the optimizer's 1,896 and pays 30.7 s of tail latency against 0.63 s, a factor of 49. In a paper whose central cost is tail latency, an unconstrained baseline winning on errors alone is not a recommendation.

Two assumptions made earlier do not survive the move. Conformal coverage, which held between 0.883 and 0.898 across the nine synthetic conditions, falls to 0.816 under normal connectivity on real data and 0.745 under low bandwidth, against a nominal 0.90. Calibration and test windows here come from different disasters, so the exchangeability Proposition 2 assumes fails within a condition and not only across conditions; per-condition recalibration cannot repair that. The spread across the three real conditions is itself 7.1 points, so a condition-dependent component remains that this explanation does not account for.

The byte model of Section V-B is contradicted by measurement. Real images have a median of 103,142.5 bytes against an assumed 220,000 to 620,000, and every report in the corpus is smaller than the smallest summary the method defines: real text runs 39 to 181 bytes against an assumed 350 to 1,800 and a summary of 184 to 224. Sending a report verbatim is therefore cheaper than summarising it, and the summarise-or-send decision is vacuous for that modality on this corpus. Of the three failures this is the one most easily avoided: byte sizes are measurable before a budget is designed around them.□

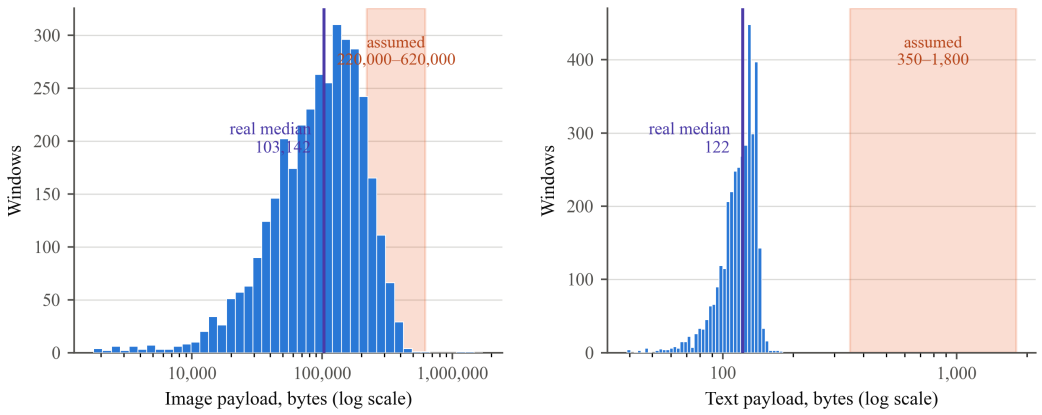


Fig. 9. Measured payload sizes against the ranges assumed in Section V-B, shaded. The image distribution lies mostly below its assumed band; the text distribution does not overlap its assumed band at all.□

VII. What the Evidence Supports

One conclusion holds across both workloads, and it is the one this paper asks to be carried forward: the value of this formulation is not in its solver. On the synthetic benchmark a simple rule matched or beat the exact optimizer in every condition: a byte-matched fixed rule under bandwidth limits, a constant wait threshold under delayed imagery, the fixed rule with the admissibility filter under misaligned reports, and a greedy uncertainty ranking when the shared budget binds. On real data the separation is sharper still: a greedy heuristic over the proposed utility returned the same allocation as the exact dynamic program

in every condition and at every seed, identical in errors, bytes and tail latency. The formulation still earns its place, but for a different reason than we expected. It identifies which quantities matter — the forecast delay, the value of the evidence already present, the incident metadata — and it supplies the threshold that turns them into a rule. The solver that consumes those quantities never reliably beat a greedy alternative on the same utility.

What does not hold across both workloads is the ranking itself. On the synthetic benchmark the fixed rule matched or beat the optimizer in all nine conditions; on real data the optimizer is ahead of that same rule in 15 of 21 comparisons over seven event splits, and a single-modality rule which cannot vary across those conditions beats them both. Which simple rule wins is therefore a property of the workload rather than of the method, and the instruction that follows is to derive and test the rule against your own held-out data rather than to inherit ours. That is a weaker claim than the synthetic results alone would license. It is the claim the evidence carries, and it is why the consequences below are stated as checks to run rather than as a policy to adopt.

Six consequences follow for anyone building such a system.

• Derive the candidates — a byte-profile fixed rule, the constant wait threshold from calibration data, and a greedy uncertainty ranking with a summary floor where a shared budget binds — and choose between them on your own held-out events. Which one wins changed between the two workloads in this paper, so the choice does not transfer; the derivation does. Keep the exact optimizer only if it beats all three.

• Before optimizing a value function, check what it prices at zero. The tight-budget failure was not a solver bug; exact optimization simply exploited a misspecification that a greedy rule was too blunt to find.

• Treat metadata admissibility as protection against accident, not against adversaries. Its value falls in direct proportion to an attacker's ability to forge, with no safe threshold.

• Recalibrate conformal thresholds per operating condition, and do not stop there: Section VI-L shows the guarantee also failing across events, so calibration has to span the boundary the deployment will actually cross. Size the review queue from reviewer capacity through the conflict threshold before any accuracy gain is worth discussing.

• Check whether a shared-budget method still helps when controllers cannot coordinate. The exact knapsack loses far more to decentralization than a greedy ranking does, and per-slot consensus over a degraded link is a cost this benchmark does not charge for.

• Add a held-out estimate of predictive contribution to the evidence value function. Section VI-L shows a real modality that is fresh, reliable, high quality and confident, and carries no signal for the label; no term in the formulation can distinguish it from a useful one, and fusing it costs more than every other component in the function is worth.

The value of the wait-or-act decision rests on two quantities this benchmark assumes: how quickly an early decision loses value, and how quickly evidence goes stale. Both have to come from emergency-management partners and measured networks, not from a synthetic decay constant.

VIII. Safety, Human Oversight and Governance

The method is decision support, not an autonomous responder. It does not issue evacuation orders, dispatch personnel, allocate scarce aid or determine eligibility. Reviewers receive the evidence set, the uncertainty, the conflict indicators, the reason codes and the status rule that fired, and the interface must distinguish observed evidence from model-derived fields. The governance design follows a lifecycle of defining context, measuring performance and harms, managing risks and maintaining accountable oversight [22]. Deployment requires agency authority, privacy review and cybersecurity controls; retention rules, accessibility testing and incident response; periodic recalibration, which Section VI-E shows is not optional; and a staffing plan for the queue measured in Section VI-F. Raw transmission must be policy-gated. Integrity digests support traceability but do not validate the truthfulness of a source, and Section VI-D shows what happens when a source is actively hostile.

IX. Limitations and Threats to Validity

The verification data are synthetic and well separated, which explains error rates below 1% for the stronger methods and makes the conformal sets nearly binary. Byte sizes, delay distributions, forecast errors and the rate at which evidence ages are assumptions, and the conclusions about fixed rules and waiting depend on them; Section VI-B shows the accuracy result under delay disappearing entirely when evidence does not age. Section VI-L removes part of this limitation, but a smaller part than the phrase 'real data' suggests. It replaces the generated workload with CrisisMMD, and against the fixed rule the comparison reverses rather than strengthens: the optimizer wins under two of the three conditions that carry over. What stays synthetic after the move is the link, the budgets, the loss model, the slot structure, the sensor modality, which has no counterpart in that corpus and is supplied by a fixed-seed placeholder, and severity, which is held constant because no field independent of the label exists. Six of the nine conditions and three of the eleven methods do not carry over at all, so the section is a reduced comparison and not a repetition of the benchmark. The link model is the sharpest limitation for this journal's readership, and Sections VI-H and VI-I narrow it rather than removing it. Loss and retransmission are now modelled, and capacity is replayed from third-party measurements rather than assumed, which is what lets us show that the constant-rate idealisation is optimistic by up to a factor of 8.7. What remains absent is a real protocol stack: there is no MAC layer, no routing, no contention between nodes, no congestion control, and no bidirectional traffic. Section VI-K measures decision time and peak allocation on one workstation; hardware latency on an edge device and energy in joules were not measured. We present the link model as an abstraction of the coupling between payload size, a shared serving order, loss and decision latency, and not as an emulation. Section VI-J relaxes the single-controller assumption but does so by static partitioning; it does not model controllers that exchange partial state, negotiate a budget, or fail independently, and the cost of any such coordination protocol is not charged to the methods that would need it.

Two measurements this study cannot make are worth naming rather than leaving implicit, because both bear directly on a deployment claim and neither is a matter of effort. There is no hardware testbed: every result here comes from simulation on a workstation, with no edge devices, no radios and no field deployment, so nothing is established about thermal behaviour, contention with other traffic on a shared device, or the failure modes of real hardware. And there is no energy measurement in joules: Section VI-K reports wall-clock time and peak allocation, which are honest proxies for computational cost, but converting them into energy would require instrumented hardware and a power model this study does not have. We report the timings and decline to infer joules from them. Both gaps need physical access, and they bound what any reader should take from this work about deployment on constrained devices.

Policy coefficients and thresholds were set by hand. The latency penalty was changed from an earlier version for a stated design reason and is accompanied by a sweep, but the protocol was not pre-registered. The information-matched baselines were added after an audit; the wait threshold was fixed on calibration events and the review settings were not changed. The

admissibility test favours the filter by construction (Section VI-C), and the adversarial sweep models metadata substitution only. The contested-budget difficulty was chosen from baselines on calibration events by a rule fixed in advance, but the budget grid and the competitor were chosen by the authors, and both protocols are timestamped only within the package. The severity estimator is weak, so severity-dependent behaviour is barely exercised. Five seeds and four test events per seed limit statistical resolution and several paired intervals include zero. On the real-data side the seeds vary model fitting only, which is why Section VI-L rotates the event split and states its comparisons over 21 rotations rather than three conditions of one split; what the rotation cannot do is add an eighth disaster, so the seven assignments reuse the same events and are not independent of one another. Social-media and sensor sources carry participation and placement bias, and reliability estimates can encode historical bias.

The claims are therefore limited to the formulation, its stated properties, and reproducible behaviour under stated assumptions. Improved disaster outcomes cannot be claimed without event-aligned real data such as CrisisMMD [4], single-modality resources such as xBD, FloodNet or Sen1Floods11 [1–3], measured network traces, and prospective studies with emergency-management partners.

X. Reproducibility

The package contains the source, an environment lock, the configuration, event split identifiers, calibration temperatures and thresholds, the derived wait thresholds, seeds and integrity tests. It also contains record-level predictions for every method and condition, provenance envelopes for every proposed decision, the two frozen contested-budget protocols with their SHA-256 digests, and scripts that regenerate every table and figure in this paper without manual copying.

On one CPU architecture, reruns reproduce every output file byte for byte, and this holds across Python 3.12 to 3.14 with the corresponding NumPy, pandas and scikit-learn releases. It does not hold across architectures: the generator's Beta and Dirichlet draws reach platform libm, so about 10% of source observation vectors differ in their last bits between Linux x86_64 and macOS arm64. Every reported quantity is written at six significant figures or fewer and is unaffected; the provenance digest is taken over the observation rounded to nine decimals so that it is stable across architectures too. That claim covers the synthetic pipeline. The real-data pipeline of Section VI-L additionally depends on a pretrained encoder and on PyTorch, whose versions its manifest does not pin and whose numerical behaviour we have not tested across platforms, so we do not claim byte-for-byte reproduction of the derived records. What is recorded instead is a digest of those records, the encoder identifier, the split, the label mapping and the source URL, in data/crisismmd/provenance.json; the corpus itself is obtained from its authors and is not redistributed here.

An earlier version of this prototype contained design errors that produced several results by construction: a feature action scored with cloud output it had never transmitted, a severity value drawn from the label, a budget applied to only one method, and a delay penalty that exceeded the maximum achievable value so that images were always held. Those results are superseded and the corrections are listed in the package changelog. We mention them here because a reader assessing a validation paper is entitled to know that the validation has already caught things, and what they were.

XI. Conclusion

We formulated disaster evidence transmission as a shared-budget knapsack across concurrent incidents, in which waiting for late evidence is an explicit decision with a delay threshold and admissibility is checked before any evidence is valued or fused, and carried every decision through to a conformal set, an operational status and a provenance record. Controlled verification shows the formulation behaves as specified and that, in this benchmark, the simple rules it generates match the optimizer it justifies. A constant wait threshold cuts tail latency at least as much as the policy when evidence arrives late; the admissibility filter alone keeps misaligned reports out of fusion; a fixed rule suffices when bandwidth is the only constraint; and when a shared budget binds, a pre-registered test found no benefit from exact optimization. Two results bear directly on trustworthiness in contested settings: the admissibility filter's protection falls to nothing in proportion to an adversary's ability to forge metadata, and the conformal guarantee does not transfer across operating conditions. Section VI-L then removed the largest qualification on all of this by running a reduced version of the comparison on real disaster data, and it complicated the conclusion rather than confirming it. Against the fixed rule the optimizer is ahead in 15 of 21 comparisons across seven event splits, and in every split under normal connectivity, reversing the synthetic verdict in direction while leaving the outcome in any single condition dependent on which disasters are held out. A simple rule sufficing is therefore workload-dependent rather than general. What beats both on errors is a single-modality rule that cannot vary across those conditions and is among the worst methods on the synthetic benchmark, and it wins because one real modality carries no signal for the label while every term in the value function rates that modality well. Two assumptions made earlier did not survive: the conformal guarantee fails within a condition once calibration and test windows come from different disasters, and the byte model is contradicted by measurement. One conclusion survives both workloads and is what we ask to be carried forward: the formulation earns its place by naming the quantities that matter and supplying the threshold that turns them into a rule, and not by the solver that consumes them, which never reliably beat a greedy alternative on either workload. Which simple rule to deploy does not transfer, and has to be derived against the data a system will actually see. What remains to be established is field value under a real protocol stack and an operational deployment; the capacity traces of Section VI-I are measured, but nothing here runs on a radio.

Appendix: Supplementary Results

TABLE XVII

Proposed (P), fixed rule (FR) and transmit all (TA) across every condition. Errors out of 720; bytes per window; 95th-percentile modelled latency in seconds.

Condition	Err P	Err FR	Err TA	Bytes P	Bytes FR	p95 P	p95 FR
Normal	4.6	4.4	4.4	45,801	1,854	0.57	0.09
Low bandwidth	4.4	4.2	4.4	3,590	1,854	0.92	0.59
Delayed imagery	9.2	17.4	15.2	2,382	1,854	354.58	1,878.85
Outage	4.2	4.4	4.4	3,678	1,854	13.87	13.71
Image missing	12.2	12.2	12.2	1,630	1,630	0.19	0.19
Text missing	12.4	13.2	12.2	2,608	784	0.21	0.16
Noisy text	6.0	6.0	6.4	3,678	1,854	0.24	0.19
Contradictory	12.4	12.4	13.0	3,678	1,854	0.24	0.19
Misaligned	8.8	20.2	18.6	3,436	1,854	0.23	0.19

TABLE XVIII

Every ablation shipped in the package, each removing the component from both policy and fusion. Positive values mean the component helped. Severity is

replaced by its training mean. Only quality and the latency penalty show an effect distinguishable from zero; admissibility's point estimate is large but its interval includes zero at five seeds.

Component (condition)	Δ errors, ablated minus full [95% CI]
Severity (low bandwidth)	+0.0 [0.0, 0.0]
Severity (normal)	+0.0 [0.0, 0.0]
Novelty (contradictory)	+0.0 [0.0, 0.0]
Reliability (contradictory)	+2.0 [-0.9, 4.9]
Reliability (noisy text)	-0.6 [-1.7, 0.5]
Quality (noisy text)	+1.4 [0.3, 2.5]
Quality (contradictory)	+1.0 [0.1, 1.9]
Recency (delayed)	-0.6 [-1.7, 0.5]
Latency penalty (delayed)	+6.0 [2.8, 9.2]
Novelty (noisy text)	+0.0 [0.0, 0.0]
Admissibility (misaligned)	+8.4 [-0.5, 17.3]

Data Availability

The complete package accompanying this paper contains the reference implementation, the synthetic generator, the configuration, record-level predictions for every method and condition, provenance envelopes, aggregate and paired tables, ablations, the latency-penalty, wait-threshold and adversarial sweeps, the review-rule sensitivity analysis, the two frozen contested-budget protocols, and the scripts that produce every table and figure in this paper. The package is maintained at https://github.com/bdata-arindam/Edge. [A dataset DOI for the deposited release of that repository to be added before submission.] The synthetic verification uses generated records and contains no personal data. Section VI-L uses CrisisMMD v2.0, a third-party corpus of public social-media posts and their attached photographs, obtained from its authors under their terms and not redistributed with this package; arrival times are recovered from the public post identifiers. What we ship is derived features, byte counts and labels, not the posts or images themselves. Any operational use of field reports or agency data, and any republication of this corpus, would require institutional, legal and privacy review.

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